

Javascript

Loops, dialog boxes

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Loops</title>
  </head>
  <body>
    <h1>Loops</h1>
    <script>
      // console.log(1);
      // console.log(2);
      // console.log(3);

      // no of lines increase => console.log()

      // let table =3;
      // for(let i=1; i<=10; i++){
      //   console.log("3 * ", i, "=", table *i);
      // }
      // for loop => iteration count
      // while loop => doesn't know iteration count , condition

      // let number = 4567;
      // let result = 0;
      // 7654

      // while (number > 0) {
      //   let lastNumber = number % 10;

      //   result = result * 10 + lastNumber;
      //   number /= 10;
      // }

      // console.log(result);

      // for (let i = 0; i < 5; i++) {
      //   for (let j = 0; j < 2; j++) {
      //     console.log(i, "+", j, "=", i + j);
      //   }
      // }

      // let x = 1;
      // do {
```



```
</body>  
</html>
```

Theory Notes

1. Loops

- **For loop** → used when the number of iterations is known.

```
for(let i=1; i<=10; i++){ console.log(i); }
```

- **While loop** → used when the number of iterations is unknown, runs until condition is false.

```
let i=1; while(i<=5){ console.log(i); i++; }
```

- **Do-while loop** → executes at least once, then checks condition.

```
let i=1; do{ console.log(i); i++; } while(i<=5);
```

- **Nested loops** → loops inside loops, useful for tables or grids.
- **Break** → exits loop immediately.
- **Continue** → skips current iteration and moves to next.

2. User Interaction

- **prompt()** → takes input from user.
- **alert()** → displays a message box.
- **confirm()** → asks for confirmation (returns true/false).

3. Common Use Cases

- Generating multiplication tables.
- Reversing numbers.
- Iterating arrays.
- Validating user input.
- Creating menus with `switch` and loops.

Practice Tasks

Short Answer

1. What is the difference between `break` and `continue`?
2. Why does a do-while loop always run at least once?
3. How does `prompt()` differ from `alert()`?
4. When would you use a while loop instead of a for loop?

Coding Exercises

1. Write a program to print the multiplication table of 5 using a for loop.
2. Reverse a number using a while loop.
3. Use a do-while loop to print numbers from 1 to 5.
4. Create a program that asks for age using `prompt()` and alerts eligibility for voting.
5. Use `confirm()` to ask if the user wants to delete a file, then show appropriate alert.
6. Write a nested loop to print a 3x3 grid of numbers.
7. Demonstrate skipping even numbers using `continue` in a loop.